

Chapter 20 - FHRP (First Hop Redundancy Protocol) + HSRP + VRRP + GLBP



What is FHRP?

FHRP stands for **First Hop Redundancy Protocol**.

It is a group of protocols that protect the **default gateway** in a network.

👉 If one router fails, another router automatically takes over.

👉 Users do NOT lose internet or network connectivity.

Why is it called "First Hop"?

When a PC sends traffic outside its network:

PC → Default Gateway → Next Network

The **default gateway is the first hop**.

If that router fails ❌

Even if internet is working ✅

Users still lose connectivity.

FHRP solves this problem 💡



Core Concept of FHRP

Multiple routers share:

✓ One **Virtual IP Address**

✓ One **Virtual MAC Address**

The PC only knows the Virtual IP.

Whichever router is Active responds.

Real-Life Example

Think of it like:

 Bank with 2 managers

One is Active

Other is Backup

If one manager leaves, second immediately handles work.

Customers don't notice any issue.

That is FHRP 🙌

Types of FHRP Protocols

Protocol	Full Form	Standard	Special Feature
HSRP	Hot Standby Router Protocol	Cisco Proprietary	Redundancy
VRRP	Virtual Router Redundancy Protocol	Open Standard	Multi-vendor
GLBP	Gateway Load Balancing Protocol	Cisco Proprietary	Load balancing

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HSRP – Hot Standby Router Protocol

Protocol: Hot Standby Router Protocol

◆ Features

- ✓ Cisco proprietary
 - ✓ Active / Standby model
 - ✓ Uses Virtual IP
 - ✓ Uses Virtual MAC
 - ✓ Group based configuration
-



Multicast Addresses

HSRP v1 → 224.0.0.2

HSRP v2 → 224.0.0.102



Virtual MAC Addresses

HSRP v1 → 0000.0c07.acXX

HSRP v2 → 0000.0c9f.fXXX

(XX = Group Number)



Working Model

Role	Function
Active	Forwards traffic
Standby Backup	
Others	Listen only

✂ HSRP Configuration (Cisco)

```
interface g0/0
ip address 172.16.0.253 255.255.255.0
standby 1 ip 172.16.0.252
standby 1 priority 110
standby 1 preempt
```



Important Commands

Command	Purpose
show standby brief	Quick status
show standby	Detailed info

Command

Purpose

show standby interface g0/0 Interface specific

✓ Successful Configuration Signs

- ✓ One router = Active
 - ✓ One router = Standby
 - ✓ Same Virtual IP
 - ✓ No continuous state change
-



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VRRP – Virtual Router Redundancy Protocol

Protocol: Virtual Router Redundancy Protocol

◆ Features

- ✓ Open standard (Cisco, Juniper, MikroTik, etc.)
 - ✓ Master / Backup model
 - ✓ Preempt enabled by default
-



Multicast Address

224.0.0.18



Virtual MAC Format

0000.5e00.01XX

If group > 99 → convert to HEX

Example:

200 → C8

MAC → 0000.5e00.01c8

🔧 Basic VRRP Config

```
interface g0/1
ip address 192.168.1.3 255.255.255.0
vrrp 1 ip 192.168.1.1
vrrp 1 priority 120
vrrp 1 preempt
```

🔧 📡 GLBP – Gateway Load Balancing Protocol

Protocol: Gateway Load Balancing Protocol

◆ Features

- ✓ Cisco proprietary
 - ✓ Supports redundancy + load balancing
 - ✓ Multiple routers forward traffic simultaneously
 - ✓ Multiple Virtual MACs
-

🔧 GLBP Config

```
interface g0/1
ip address 192.168.1.4 255.255.255.0
glbp 1 ip 192.168.1.1
glbp 1 priority 110
glbp 1 preempt
```

📊 Key Differences

Feature	HSRP	VRRP	GLBP
Standard	Cisco	Open	Cisco
Model	Active/Standby Master/Backup AVG/AVF		
Load Balancing	No	No	Yes

Feature	HSRP	VRRP	GLBP
Preempt	Manual	Default	Manual

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FULL LAB – HSRP Implementation

Lab Topology Design

You need:

- ✓ 2 Routers (R1, R2)
- ✓ 4 Switches
- ✓ 4 PCs

Connections:

R1 → SW1,SW2 & 2 PC

R2 → SW3,SW4 & 2 PC

Switch cross connections for redundancy.

IP Address Plan

Network: 172.16.0.0/24

Device	IP
--------	----

R1	172.16.0.253
----	--------------

R2	172.16.0.254
----	--------------

Virtual IP 172.16.0.252

PCs	172.16.0.1 – 172.16.0.4
-----	-------------------------

Subnet Mask: 255.255.255.0

Default Gateway: 172.16.0.252

🔧 R1 Configuration

```
hostname R1
interface g0/0
ip address 172.16.0.253 255.255.255.0
standby 1 ip 172.16.0.252
standby 1 priority 110
standby 1 preempt
no shutdown
```

🔧 R2 Configuration

```
hostname R2
interface g0/0
ip address 172.16.0.254 255.255.255.0
standby 1 ip 172.16.0.252
standby 1 priority 150
standby 1 preempt
no shutdown
```

Higher priority (150) → R2 becomes Active 🏰

💻 PC Configuration

IP: 172.16.0.1 – 172.16.0.4
Mask: 255.255.255.0
Gateway: 172.16.0.252

Verification Commands

1 Quick Check

show standby brief

2 Detailed Check

show standby

Shows:

- ✓ Active Router
 - ✓ Standby Router
 - ✓ Priority
 - ✓ Timers
 - ✓ Virtual MAC
 - ✓ Preemption
-

Failover Test

Shutdown Active router:

```
interface g0/0  
shutdown
```

Standby should immediately become Active.

Users should not lose connectivity.

That means lab is successful 🎉

Important Concepts for Exam

- ✓ Understand Virtual IP logic
 - ✓ Priority decides Active
 - ✓ Preempt allows takeover
 - ✓ Always verify with show commands
 - ✓ Gateway on PC must be Virtual IP
-

CHAPTER SUMMARY

1 What is FHRP?

FHRP (First Hop Redundancy Protocol) is a group of protocols that provide **default gateway redundancy**.

- 📌 If one router fails, another router automatically takes over.
- 📌 End users do NOT notice any downtime.

Why “First Hop”?

When a PC sends traffic outside its network:

PC → Default Gateway → Next Network 

The **default gateway** is the *first hop*.

If gateway fails ❌

Even if internet is working ✅

Users lose connectivity.

👉 FHRP solves this problem.

Core Concept of FHRP

Multiple routers share:

✓ One **Virtual IP Address**

✓ One **Virtual MAC Address**

PC default gateway = **Virtual IP**

Whichever router is Active responds 🏰

Types of FHRP

Protocol Full Form		Standard	Special Feature
HSRP	Hot Standby Router Protocol	Cisco Proprietary	Redundancy
VRRP	Virtual Router Redundancy Protocol	Open Standard	Multi-vendor
GLBP	Gateway Load Balancing Protocol	Cisco Proprietary	Load Balancing

2 HSRP – Hot Standby Router Protocol

◆ Features

- ✓ Cisco proprietary
 - ✓ Active / Standby model
 - ✓ Uses Virtual IP & MAC
 - ✓ Group based
 - ✓ No load balancing
-

Multicast Addresses

- HSRP v1 → 224.0.0.2
 - HSRP v2 → 224.0.0.102
-

Virtual MAC Format

- v1 → 0000.0c07.acXX
- v2 → 0000.0c9f.fXXX

(XX = Group Number)

Roles

Role	Function
------	----------

Active	Forwards traffic
--------	------------------

Standby Backup	
----------------	--

Others	Listen only
--------	-------------

Important HSRP Concepts

- Priority decides Active (default = 100)
 - Higher priority wins 
 - Preempt must be enabled manually
 - No preemption by default
 - Timers:
 - Hello = 3 sec
 - Hold = 10 sec
-

**3**

VRRP – Virtual Router Redundancy Protocol

◆ Features

- ✓ Open standard (Cisco, Juniper, MikroTik etc.)
 - ✓ Master / Backup model
 - ✓ Preempt enabled by default
 - ✓ Faster failover
-



Multicast Address

224.0.0.18

Virtual MAC Format

0000.5e00.01XX

If group > 99 → convert to HEX

Example:

Group 200 → C8

MAC → 0000.5e00.01c8

Important Differences from HSRP

- Master instead of Active
 - Preempt enabled by default
 - Default priority = 100
 - Real IP of Master can be Virtual IP
-

GLBP – Gateway Load Balancing Protocol

◆ Features

- ✓ Cisco proprietary
 - ✓ Redundancy + Load Balancing
 - ✓ Multiple routers forward traffic
 - ✓ Multiple Virtual MACs
-

Roles

Role Meaning

AVG Active Virtual Gateway

AVF Active Virtual Forwarder

AVG assigns virtual MACs to AVFs.

Load Balancing Methods

- Round Robin
 - Weighted
 - Host-dependent
-

HSRP Lab Summary

Network: 172.16.0.0/24

Virtual IP: 172.16.0.252

R1: 172.16.0.253

R2: 172.16.0.254

PC Gateway = 172.16.0.252

Higher priority → Active 

Failover test:

Shutdown Active → Standby becomes Active instantly 

FINAL CONCLUSION

FHRP ensures:

- ✓ Gateway redundancy
- ✓ High availability
- ✓ Zero user downtime
- ✓ Business continuity

HSRP = Redundancy (Cisco)

VRRP = Redundancy (Multi-vendor)

GLBP = Redundancy + Load balancing

 In real enterprise networks, FHRP is mandatory.

Q & A

1 What is FHRP?

FHRP is a group of protocols that provide redundancy for the default gateway so that if one router fails, another takes over automatically.

2 Why is FHRP required?

Without FHRP:

If default gateway fails → entire LAN loses connectivity ❌

With FHRP:

Backup router takes over automatically ✅

3 What is Virtual IP?

Virtual IP is the shared IP address configured as default gateway on PCs.

It is not a physical interface IP.

4 What is Virtual MAC?

Virtual MAC is the MAC address associated with the Virtual IP.

It prevents ARP changes during failover.

5 Difference between HSRP and VRRP?

HSRP

VRRP

Cisco proprietary Open standard

Active/Standby Master/Backup

Preempt manual Preempt default

HSRP

VRRP

Uses 224.0.0.2 Uses 224.0.0.18

6 What is GLBP?

GLBP provides redundancy AND load balancing.

Multiple routers forward traffic simultaneously.

7 How does HSRP elect Active router?

1. Highest priority wins
2. If tie → Highest IP address wins

Default priority = 100

8 What is Preempt?

Preempt allows higher priority router to take back Active role when it comes back online.

Without preempt → No takeover.

9 What are default HSRP timers?

Hello = 3 sec

Hold = 10 sec

10 What is HSRP group?

Group number identifies HSRP instance.

Different VLANs can use different groups.

1 1 Can HSRP perform load balancing?

No ❌

Only one Active forwards traffic.

1 2 How VRRP is better than HSRP?

- Open standard
 - Faster failover
 - Preempt default
-

1 3 What happens if Active router fails?

Standby immediately becomes Active ⚡

Virtual MAC remains same → No ARP update needed.

1 4 What happens if priorities are equal?

Highest IP address becomes Active.

1 5 Why Virtual MAC is important?

If MAC changes during failover → ARP update required → delay occurs.

Virtual MAC ensures seamless failover 🚀

1 6 How GLBP provides load balancing?

GLBP assigns different Virtual MAC addresses to different routers.

Different PCs get different ARP replies.

Traffic distributed automatically.

1 7 What is AVG in GLBP?

Active Virtual Gateway.

Controls MAC assignment.

1 8 What is AVF?

Active Virtual Forwarder.

Forwards traffic using assigned Virtual MAC.

1 9 How to verify HSRP?

show standby brief

show standby

2 0 Troubleshooting: Standby not becoming Active?

Check:

✓ Priority

✓ Preempt

✓ Timers

✓ Interface status

✓ Same group number

2 1 Configure HSRP on two routers.

(Explain step-by-step like lab section.)

2 2 How to test failover?

Shutdown Active interface.

Check if Standby becomes Active.

Ping external IP to confirm.

2 3 What if Virtual IP not configured on PC?

FHRP won't work.

PC must use Virtual IP as gateway.

2 4 Can we run multiple HSRP groups?

Yes 

Used for load sharing between VLANs.

2 5 What is object tracking in HSRP?

If WAN link fails, router reduces priority.

Backup router becomes Active automatically.

Improves intelligent failover 

2 6 Difference between GLBP and HSRP in real network?

HSRP → Only redundancy

GLBP → Redundancy + Traffic distribution

GLBP preferred when load balancing required.

2 7 Why businesses use FHRP?

Because downtime costs money 

FHRP ensures high availability.
